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The Editors
Physical Chemistry Chemical Physics

Dear Editors,

On behalf of my co-author, I am submitting the manuscript **"What a radical-pair quantum reservoir would require: readable capacity, no quantum advantage, and a nuclear-register criterion"** for consideration as a research article in *PCCP*.

This is a re-submission, and I should say plainly what happened. It supersedes manuscript **CP-ART-06-2026-002404**, which we asked to withdraw during review after discovering ourselves that one of its central numerical results was an artifact of our simulation protocol rather than a property of the spin chemistry. The present manuscript is the corrected and substantially extended analysis. Its conclusions differ from the withdrawn version in three places, and in each case the corrected answer is the more interesting one. We are grateful for the opportunity to put the corrected work before the journal.

The question is whether the spin chemistry of a cryptochrome-type radical pair can *process* information—posed, as it must be for a warm and noisy system, as reservoir computing rather than gate-model computation. We solve the Liouville-space Haberkorn master equation with electronic decoherence and, crucially, drive the reservoir the way a real turnover does: the nuclei are retained and a new spin-correlated pair is born each cycle, its singlet character encoding the input. Resetting a single electron instead—the natural abstraction, and the one used in the withdrawn version—destroys the electron–electron correlation and, for a separated pair, pins the singlet probability at exactly 1/4 regardless of input. That is the artifact, and correcting it changes the physics.

Three results follow. **First**, the reservoir is readable by ordinary chemistry. The time-resolved yield of one product pool gives an out-of-sample information-processing capacity of 2.0, and the nuclear polarisation the product carries away raises it to 4.7, with no spectroscopic access of any kind. Readout is not the bottleneck, as we had previously concluded. The mechanism behind that number is itself worth reporting to a physical-chemistry readership: because either electron of a newly born pair is maximally mixed for every input, spin-selective recombination is what *writes* the input into the nuclear register. In a separated pair, CIDNP is not an optional additional channel; it is the write mechanism.

Second, there is no quantum advantage. A classical echo-state network given the same number of readout features exceeds the five-spin reservoir—at the cryptochrome operating point by 49% at eight matched channels and 123% at five. Matched by physical unit rather than channel count the two are indistinguishable (4%, half a standard deviation); on no accounting is the radical pair ahead by more than its own scatter. We report this rather than argue around it.

Third, the timescale objection usually raised against this system does not survive. It rests on identifying the reservoir's clock with the microsecond radical-pair lifetime, but the memory is carried by the nuclear register, which is not consumed by recombination and departs in the diamagnetic product. Across five decades of turnover interval the memory capacity falls by only 1.6% (twelve input realisations, paired), so a 10–100 ms turnover places the horizon at 19–188 ms, inside the neural band—provided the register itself survives the pause, which we show is now the binding condition: at the geomagnetic field nuclear relaxation is in the extreme-narrowing regime, and a proton on the intact protein relaxes in milliseconds rather than seconds. What replaces the timescale argument is a falsifiable chemical criterion: the *same* nuclear register must be reused from one turnover to the next, failing which the capacity collapses to a memoryless read-back. The spin-dynamical analysis is what makes that criterion usable, because it identifies what the register actually stores: pure nuclear dephasing at any rate leaves the capacity untouched, so the memory lives in nuclear populations and not in coherence. We should add at once that this distinction, which we had expected to be the rescue, changes none of the numbers at the operating point: the geomagnetic field puts the register in extreme narrowing, where T_1 and T_2 coincide.

We should be equally plain about where that leaves the biological reading, since it is less favourable than in the withdrawn version. Predicting the relaxation time rather than assuming it—from measured flavin quadrupole couplings and standard geometry, with the calculator validated against three measured systems—shows that a rigidly protein-bound register cannot reach the neural band at all: at the geomagnetic field the system is in extreme narrowing, a proton on a 60 kDa protein relaxes in 2.4 ms, and the horizon is capped at 5.5 ms. The register must reorient at least three times faster than the whole protein for the band to open. The ^{14}N of the flavin, being quadrupolar, cannot hold a register at all. And the drive cannot be light: a single flavin absorbs once per 7.7 s in full sunlight and once per some hundreds of years inside a skull, so the regeneration must be catalytic. What survives is a specific and testable statement—catalytic regeneration on a proton register that reorients with $\tau_c \lesssim 5$ ns, at a turnover interval set by that mobility (10–22 ms at $\tau_c = 5$ ns, up to 5–1100 ms at 0.1 ns)—and both of its parameters are measurable in vitro, by relaxation dispersion and steady-state kinetics, without a neuron. We would rather submit the smaller claim with its constraints made explicit than the larger one.

The work is entirely computational. All code and numerical outputs are openly released in the `qbscreen` package, every capacity is reported out of sample with its null floor and its convergence in sample length, and the central claims are covered by an automated test suite (246 tests, all passing): twenty-nine bind the headline quantities and one hundred and twenty-eight bind the capacity and standard-deviation cells of every table, with the expected values transcribed from the manuscript rather than from the data files. A further six parse the printed cells back out of the `.tex` and check them against the data, so the suite fails if either side moves, and twelve bind the input-parameter table. Derived-time and raw-IPC columns are not yet covered. A companion study on the corresponding weak-field magnetosensing analysis is under consideration elsewhere; the two are physically independent and we are glad to share it. This manuscript is not under consideration at any other journal. The authors declare no competing interests and the study received no specific funding.

Thank you for considering this re-submission.

Yours sincerely,

Hikaru Wakaura